LIST --> Mutable

In [22]:
```
l=[]
l
```
Out[22]: `[]`

In [23]:
```
type(l)
```
Out[23]: `list`

In [24]:
```
len(l)
```
Out[24]: `0`

Append()

In [25]:
```
l.append(10) #append()--> add the value at the end of the list
l
```
Out[25]: `[10]`

In [26]:
```
len(l)
```
Out[26]: `1`

In [27]:
```
l.append(10,20) #only one argument can be appended
l
```
```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[27], line 1
----> 1 l.append(10,20) #only one argument can be appended
      2 l

TypeError: list.append() takes exactly one argument (2 given)
```

In [28]:
```
l1=[10]
l1
```
Out[28]: `[10]`

In [29]:
```
l1.append(20)
l1.append(30)
l1.append(40)
l1.append(50)
l1
```
Out[29]: `[10, 20, 30, 40, 50]`

In [30]:
```
len(l1)
```
Out[30]: `5`

In [31]:
```
l1.append(10)  #Duplicate is allowed
l1
```
Out[31]: [10, 20, 30, 40, 50, 10]

In [32]:
```
l2=[10,2.8,"mousumi",12+2j,True] #In list we can assign diffrent types of dataty
l2
```
Out[32]: [10, 2.8, 'mousumi', (12+2j), True]

In [33]:
```
l2.append([1,2,3]) #nested list --> list inside another list is nested list
l2
```
Out[33]: [10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

copy()

In [34]:
```
print(l1)
print(l2)
```
```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

In [35]:
```
l1
```
Out[35]: [10, 20, 30, 40, 50, 10]

In [36]:
```
l3=l1.copy() #to copy one list to another list
l3
```
Out[36]: [10, 20, 30, 40, 50, 10]

In [37]:
```
print(l1)
print(l2)
print(l3)
```
```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
[10, 20, 30, 40, 50, 10]
```

In [38]:
```
l1==l3
```
Out[38]: True

In [39]:
```
len(l1)==len(l3)
```
Out[39]: True

In [40]:
```
id(l1)==id(l3) #have same value but the id is not same
```
Out[40]: False

In [41]:
```
print(id(l1))  #both have diffrent id
print(id(l3))
```

```
1232003825984
1232003097600
```

clear()

In [42]:
```
print(l1)
print(l2)
print(l3)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
[10, 20, 30, 40, 50, 10]
```

In [43]:
```
l3.clear() #it clear all the value and gives empty list
```

In [44]:
```
print(l1)
print(l2)
print(l3)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
[]
```

In [45]:
```
del l3 #to delete any variable we use delete
```

In [46]:
```
l3
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[46], line 1
----> 1 l3

NameError: name 'l3' is not defined
```

In [47]:
```
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

count()

In [48]:
```
l1.count(20) #count the occurancces of variable
```

Out[48]: 1

In [49]:
```
l1.count(10)
```

Out[49]: 2

extend()

In [50]:
```
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

In [51]:
```
l1.extend(l2) #we can merge the list
l1
```

Out[51]:
```
[10, 20, 30, 40, 50, 10, 10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

In [52]:
```
print(len(l1))
print(len(l2))
```

```
12
6
```

index()

In [53]:
```
print(l1)
print(l2)
```

```
[10, 20, 30, 40, 50, 10, 10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

In [54]:
```
l1.index(20) #indexing is allowed in list
```

Out[54]:
```
1
```

In [55]:
```
l2.index(12+2j)
```

Out[55]:
```
3
```

In [56]:
```
l2.index("mousumi")
```

Out[56]:
```
2
```

In [57]:
```
l2[:]
```

Out[57]:
```
[10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
```

In [58]:
```
l2[5]
```

Out[58]:
```
[1, 2, 3]
```

In [59]:
```
l2[6]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[59], line 1
----> 1 l2[6]

IndexError: list index out of range
```

In [60]:
```
l2[:3]
```

Out[60]:
```
[10, 2.8, 'mousumi']
```

In [61]:
```
l2[3:]
```

Out[61]: [(12+2j), True, [1, 2, 3]]

In [62]:
```
l2
```

Out[62]: [10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

In [63]:
```
l2[0:5:2]
```

Out[63]: [10, 'mousumi', True]

In [64]:
```
l2[-3]
```

Out[64]: (12+2j)

In [65]:
```
l2[-3:]
```

Out[65]: [(12+2j), True, [1, 2, 3]]

In [66]:
```
l2[::-1]
```

Out[66]: [[1, 2, 3], True, (12+2j), 'mousumi', 2.8, 10]

In [67]:
```
l2[::-2]
```

Out[67]: [[1, 2, 3], (12+2j), 2.8]

In [68]:
```
l2
```

Out[68]: [10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

In [69]:
```
l2[2]
```

Out[69]: 'mousumi'

In [76]:
```
print(l2[2][0]) #nested index--> index within another index
print(l2[2][1])
print(l2[2][2])
print(l2[2][3])
print(l2[2][4])
print(l2[2][5])
print(l2[2][6])
print(l2[2][7])
```

```
m
o
u
s
u
m
i
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[76], line 8
      6 print(l2[2][5])
      7 print(l2[2][6])
----> 8 print(l2[2][7])

IndexError: string index out of range
```

In [71]:
```
l2
```

Out[71]: [10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

In [72]:
```
l2[0]=100  #mutable-->changable (list is mutable)
l2
```

Out[72]: [100, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

In [73]:
```
l2
```

Out[73]: [100, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

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insert

In [74]:
```
l3=[10,20,30]
l3
```

Out[74]: [10, 20, 30]

In [75]:
```
l3[0]=100
l3
```

Out[75]: [100, 20, 30]

In [77]:
```
l3.insert(2,25)
l3
```

Out[77]: [100, 20, 25, 30]

In [78]:
```
l3.insert(2,[1,2,3])
l3
```

Out[78]: [100, 20, [1, 2, 3], 25, 30]

pop() -->remove value based on indexing

In [79]:
```
l3.pop()
```

Out[79]: 30

In [80]:
```
l3
```

Out[80]: [100, 20, [1, 2, 3], 25]

In [81]:
```
l2
```

Out[81]: [100, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]

In [82]:
```
l2.pop()
```

Out[82]: [1, 2, 3]

In [83]:
```
l2.pop(-2)
```

Out[83]: (12+2j)

In [84]:
```
l2
```

Out[84]: [100, 2.8, 'mousumi', True]

remove() --> direct remove value not indexing

In [90]:
```
l2.remove(2.8)
```

In [91]:
```
l2
```

Out[91]: [100, 'mousumi']

reverse()

In [92]:
```
print(l1)
print(l2)
print(l3)
```

```
[10, 20, 30, 40, 50, 10, 10, 2.8, 'mousumi', (12+2j), True, [1, 2, 3]]
[100, 'mousumi']
[100, 20, [1, 2, 3], 25]
```

In [95]:
```
l2.reverse()
l2
```

Out[95]: ['mousumi', 100]

In [96]:
```
l3[::-1]
```

Out[96]: [25, [1, 2, 3], 20, 100]

sort() --> bydefault it gives ascending sorting

--> it only works when we have similar datatype

In [97]:
```
l4=[3,300,23,200,1000,10]
l4
```

Out[97]: [3, 300, 23, 200, 1000, 10]

In [98]:
```
l4.sort() #bydefault it is reverse=False i.e, descending sort
l4
```

Out[98]: [3, 10, 23, 200, 300, 1000]

In [99]:
```
l4.sort( reverse=True)
l4
```

Out[99]: [1000, 300, 200, 23, 10, 3]

In [100…
```
l3.sort()
l3
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[100], line 1
----> 1 l3.sort()
      2 l3

TypeError: '<' not supported between instances of 'list' and 'int'
```

list membership

In [101…
```
l4
```

Out[101… [1000, 300, 200, 23, 10, 3]

In [ ]:

In [102…
```
10 in l4
```

Out[102… True

In [103…
```
100 in l4
```

Out[103… False

All and Any --> in list if we have 0 value ALL will print false , else true

In [104…
```
l4
```

Out[104… [1000, 300, 200, 23, 10, 3]

In [105…
```
all(l4)
```

Out[105… True

In [106…
```
any(l4)
```

Out[106… True

In [107…
```
l4.append(0)
l4
```
Out[107…
```
[1000, 300, 200, 23, 10, 3, 0]
```
In [108…
```
all(l4)
```
Out[108…
```
False
```
In [109…
```
any(l4)
```
Out[109…
```
True
```

PRACTICE (17-08-26)

strip -->Removes whitespace

lstrip --> Removes whitespace from left side only

rstrip --> Removes whitespace from right side only

In [10]:
```
txt = "   abc def ghi   "

txt.lstrip()
```
Out[10]:
```
'abc def ghi   '
```
In [6]:
```
txt = "  abc def ghi  "
txt.strip()
```
Out[6]:
```
'abc def ghi'
```
In [7]:
```
txt = "   abc def ghi   "

txt.rstrip()
```
Out[7]:
```
'   abc def ghi'
```

escape character (\ )

In [8]:
```
#Using double quotes in the string is not allowed.
mystr = "My favourite TV Series is "Game of Thrones""
```
```
  Cell In[8], line 2
    mystr = "My favourite TV Series is "Game of Thrones""
                                         ^
SyntaxError: invalid syntax
```
In [9]:
```
mystr = "My favourite series is \"Game of Thrones\""
```

```
print(mystr)
```

```
My favourite series is "Game of Thrones"
```

List question practice

In [11]:
```
list1 = [] # Empty List
```

In [12]:
```
print(type(list1))
```

```
<class 'list'>
```

In [13]:
```
list2 = [10,30,60] # List of integers numbers
```

In [14]:
```
list3 = [10.77,30.66,60.89] # List of float numbers
```

In [15]:
```
list4 = ['one','two' , "three"] # List of strings
```

In [16]:
```
list5 = ['Asif', 25 ,[50, 100],[150, 90]] # Nested Lists
```

In [17]:
```
list6 = [100, 'Asif', 17.765] # List of mixed data types
```

In [18]:
```
list7 = ['Asif', 25 ,[50, 100],[150, 90] , {'John' , 'David'}]
```

In [19]:
```
len(list6) #Length of list
```

Out[19]:
```
3
```

list indexing

In [20]:
```
list2[0] # Retreive first element of the list
```

Out[20]:
```
10
```

In [21]:
```
list4[0] # Retreive first element of the list
```

Out[21]:
```
'one'
```

In [22]:
```
list4[0][0] # Nested indexing - Access the first character of the first list ele
```

Out[22]:
```
'o'
```

In [23]:
```
list4[-1] # Last item of the list
```

Out[23]:
```
'three'
```

In [24]:
```
list5[-1] # Last item of the list
```

Out[24]:
```
[150, 90]
```

list slicing

In [25]:
```
mylist = ['one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight']
```

In [26]:
```
mylist[0:3] # Return all items from 0th to 3rd index location excluding the item
```

Out[26]: ['one', 'two', 'three']

In [27]:
```
mylist[2:5] # List all items from 2nd to 5th index location excluding the item a
```

Out[27]: ['three', 'four', 'five']

In [28]:
```
mylist[:3] # Return first three items
```

Out[28]: ['one', 'two', 'three']

In [29]:
```
mylist[:2] # Return first two items
```

Out[29]: ['one', 'two']

In [30]:
```
mylist[-3:] # Return last three items
```

Out[30]: ['six', 'seven', 'eight']

In [31]:
```
mylist[-2:] # Return last two items
```

Out[31]: ['seven', 'eight']

In [32]:
```
mylist[-1] # Return last item of the list
```

Out[32]: 'eight'

In [33]:
```
mylist[:] # Return whole list
```

Out[33]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

Add,remove and change items

In [34]:
```
mylist
```

Out[34]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

In [35]:
```
mylist.append('nine') # Add an item to the end of the list
mylist
```

Out[35]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [36]:
```
mylist.insert(9,'ten') # Add item at index location 9
mylist
```

Out[36]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']

In [37]:
```
mylist.insert(1,'ONE') # Add item at index location 1
mylist
```

Out[37]:
```
['one',
 'ONE',
 'two',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'ten']
```

In [38]:
```
mylist.remove('ONE') # Remove item "ONE"
mylist
```

Out[38]: `['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']`

In [39]:
```
mylist.pop() # Remove last item of the list
mylist
```

Out[39]: `['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']`

In [40]:
```
mylist.pop(8) # Remove item at index location 8
mylist
```

Out[40]: `['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']`

In [41]:
```
del mylist[7] # Remove item at index location 7
mylist
```

Out[41]: `['one', 'two', 'three', 'four', 'five', 'six', 'seven']`

In [42]:
```
#changing value  of the string

mylist[0] = 1
mylist[1] = 2
mylist[2] = 3
mylist
```

Out[42]: `[1, 2, 3, 'four', 'five', 'six', 'seven']`

In [43]:
```
mylist.clear() # Empty List / Delete all items in the list
mylist
```

Out[43]: `[]`

In [44]:
```
del mylist # Delete the whole list
mylist
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[44], line 2
      1 del mylist # Delete the whole list
----> 2 mylist

NameError: name 'mylist' is not defined
```

copy list

In [45]:
```
mylist = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine'
```

In [88]:
```
mylist1 = mylist # Create a new reference "mylist1"  (= --> same box)
```

In [47]:
```
id(mylist) , id(mylist1) # The address of both mylist & mylist1 will be the same
```

Out[47]: (2113068301632, 2113068301632)

In [89]:
```
mylist2 = mylist.copy() # Create a copy of the list (.copy -> new box)
```

In [49]:
```
id(mylist2) # The address of mylist2 will be different from mylist because mylis
```

Out[49]: 2113074732864

In [51]:
```
mylist[0] = 1
```

In [52]:
```
mylist
```

Out[52]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [53]:
```
mylist1 # mylist1 will be also impacted as it is pointing to the same list
```

Out[53]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [90]:
```
mylist2 # Copy of list won't be impacted due to changes made on the original lis
```

Out[90]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

join list

In [55]:
```
list1 = ['one', 'two', 'three', 'four']
list2 = ['five', 'six', 'seven', 'eight']
```

In [56]:
```
list3 = list1 + list2 # Join two lists by '+' operator
list3
```

Out[56]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

In [57]:
```
list1.extend(list2) #Append list2 with list1
list1
```

Out[57]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

list membership

In [58]:
```
list1
```

Out[58]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

In [59]:
```
'one' in list1 # Check if 'one' exist in the list
```
Out[59]:
```
True
```
In [60]:
```
'ten' in list1 # Check if 'ten' exist in the list
```
Out[60]:
```
False
```
In [63]:
```
if 'three' in list1: # Check if 'three' exist in the list
  print('Three is present in the list')
else:
  print('Three is not present in the list')
```
```
Three is present in the list
```
In [64]:
```
if 'eleven' in list1: # Check if 'eleven' exist in the list
  print('eleven is present in the list')
else:
  print('eleven is not present in the list')
```
```
eleven is not present in the list
```

Reverse and sort list

In [66]:
```
list1
```
Out[66]:
```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```
In [67]:
```
list1.reverse() # Reverse the list
list1
```
Out[67]:
```
['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```
In [68]:
```
list1 = list1[::-1] # Reverse the list
list1
```
Out[68]:
```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```
In [69]:
```
mylist3 = [9,5,2,99,12,88,34]
mylist3.sort() # Sort list in ascending order
mylist3
```
Out[69]:
```
[2, 5, 9, 12, 34, 88, 99]
```
In [70]:
```
mylist3 = [9,5,2,99,12,88,34]
mylist3.sort(reverse=True) # Sort list in descending order
mylist3
```
Out[70]:
```
[99, 88, 34, 12, 9, 5, 2]
```
In [71]:
```
mylist4 = [88,65,33,21,11,98]
sorted(mylist4) # Returns a new sorted list and doesn't change original list
```
Out[71]:
```
[11, 21, 33, 65, 88, 98]
```
In [73]:
```
mylist4
```

Out[73]: [88, 65, 33, 21, 11, 98]

count

In [74]:
```
list10 =['one', 'two', 'three', 'four', 'one', 'one', 'two', 'three']
```

In [75]:
```
list10.count('one') # Number of times item "one" occurred in the list.
```

Out[75]: 3

In [76]:
```
list10.count('two') # Occurence of item 'two' in the list
```

Out[76]: 2

In [77]:
```
list10.count('four') #Occurence of item 'four' in the list
```

Out[77]: 1

All / Any

The all() method returns:

True - If all elements in a list are true

False - If any element in a list is false

The any() function returns True if any element in the list is True. If not, any() returns False.

In [78]:
```
L1 = [1,2,3,4,0]
```

In [79]:
```
all(L1) # Will Return false as one value is false (Value 0)
```

Out[79]: False

In [80]:
```
any(L1) # Will Return True as we have items in the list with True value
```

Out[80]: True

In [81]:
```
L2 = [1,2,3,4,True,False]
```

In [82]:
```
all(L2) # Returns false as one value is false
```

Out[82]: False

In [83]:
```
any(L2) # Will Return True as we have items in the list with True value
```
Out[83]:
```
True
```
In [84]:
```
L3 = [1,2,3,True]
```
In [85]:
```
all(L3) # Will return True as all items in the list are True
```
Out[85]:
```
True
```
In [87]:
```
any(L3)
```
Out[87]:
```
True
```
In [ ]: